

of blockchain-oriented apps. The key features of blockchain oriented software (BOS) are listed below:

- The data is replicated and stored in thousands of systems that ensure data security. Each node has a copy of the blockchain code.
- It checks the transaction requirements before processing for validation.
- It records transactions in a sequential log of interlinked blocks which are created by a consensus algorithm.
- Transactions are based on public-key cryptography.
- The OS serves as the base for BOS engineering.

How blockchain is used in the software industry

Blockchain uses a distributed database to store records to make it accessible and secure for everyone. This technology is safe and protects against unauthorised access. Built on peer-to-peer technology, it eliminates the need for a middleman in the process. It is a robust and secure technology that is used for developing software applications.

Anyone using an application developed using blockchain technology can view and add a record in the database but can't delete or modify the database record. Limited access ensures robust databases that improve data integrity as well. This technology also ensures security benefits to business-critical applications. It is used in the finance industry, where security is an essential feature as each transaction is vital in this industry.

Renowned companies are exploring opportunities to integrate blockchain technology in their database to ensure security. For instance, GE is planning to

implement such blockchain developed databases in its aviation division. Also, the company is working on software that could manage sales, tracking and record-keeping, and inventory details with high-end security, transparency and integrity.

In conventional development methods, transparency counters integrity as transparency of data increases the chances of tampering. But in blockchain technology, both exist together and are offered as an added advantage to companies.

Empowerment of newcomers in the industry

Trust and reliability are the two critical components of third-party applications that act as middlemen when it comes to financial transactions, shopping and more. These apps charge you for a business transaction done through their platform. Blockchain gives you the authority to execute a financial transaction without paying a third party. It has a peer-to-peer network with self-executing contracts that ensure payments between customers and applications.

There are new platforms such as BigchainDB and Hyperledger, which can be used by blockchain developers to gain scalability and permissions benefits. BigchainDB solves the scalability issues faced in implementing blockchain technology. Hyperledger enables the creation of private permissions for blockchains.

Companies are leveraging the benefits of blockchain with cloud based implementation. Leading companies such as Amazon, IBM and Microsoft are providing cloud based development spaces to software companies. Blockchain as a Service, or BaaS, offers various benefits to companies and helps them in creating viable solutions to cater to business needs.

The use of blockchain in a financial institution

Banks or other financial institutions have numerous divisions, which are allowed to execute transactions within the divisions. They are responsible for transacting loans, mortgages and other financial services as well. Now, banks are using blockchain technology to create their cryptocurrency based on smart contracts and IoT network features.

What are smart contracts?

Smart contracts ease the process by reducing the requirement of a middleman or a lawyer. Customers can mortgage property or purchase it by approaching a bank. The bank can initiate a smart contract, which will be accepted without involving any third party or person. Once the customer's identification and details are verified, the bank executes the payment to the seller of the property. The bank will also activate automatic repayments from the buyer's account until the loan is fully repaid. The smart contract ends after the full reimbursement of the loan, automatically. BOS could save millions every year and could lead to efficient transactions. Smart contracts also reduce the risk of fraud and mistakes in loan delivery.

Blockchain can help to overcome a lot of the challenges in the development of applications. A few things, however, remain to be addressed in the distributed system. **END** 

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